

GAR-HGNN: A Heterogeneous Graph-Based Approach to Recruitment Fraud Detection

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Abstract

In this paper, we propose a gated attention and dual feature enhancement method GAR-HGNN based on heterogeneous graph to address the problem of low recall and F1 value in recruitment fraud detection. By dynamically adjusting the fusion weights of text and attribute features through the gated attention mechanism, combining the labour market and criminology theories, we design industry-salary anomaly index (node attribute) and geography-industry risk density (edge weight) to enhance the graph structure risk expression, and optimize the training of imbalanced data by using the weighted loss function based on the anomaly index. Experiments show that the method achieves an accuracy of 0.9896, precision of 0.9626, recall of 0.9962, and F1 value of 0.9791 on the EMSCD dataset.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing.

Keywords

Recruitment fraud detection, heterogeneous graph neural network, gated attention mechanism, class imbalance, deep learning, anomaly index, risk density

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1 INTRODUCTION

In the digital era, recruitment platform has become a key link between job seekers and enterprises[1]. CNNIC report shows that in June 2023, China's online recruitment user scale reached 455 million, accounting for 43.5% of Internet users; Check Point report points out that in 2022, the global recruitment fraud cases increased by 37% year-on-year, with an average loss of more than 5,000 U.S. dollars, and the frequent occurrence of fraudulent behaviours, such as fictitious positions, has led to the leakage of job seekers' information and corporate reputation risk[2]. Fraudulent behaviours

such as fictitious positions occur frequently, leading to information leakage of job seekers and corporate reputation risk[2].

Recruitment fraud information is heterogeneous, containing abnormal high-frequency words, unstructured text and metadata. According to the information asymmetry theory[3], fraudsters use anomalous conditions to attract job seekers, while traditional rule-based methods are difficult to cope with dynamic patterns and have serious leakage detection[4], and machine learning models suffer from overfitting, weak recognition of few classes, and under-utilisation of graph structural information in high-dimensional sparse and class imbalance (real/fake jobs are about 20:1 scenarios[5]. Although graph neural networks (GNNs) are advantageous in modelling complex relationships[6], ordinary GNNs are difficult to integrate heterogeneous information, while heterogeneous graph neural networks (HGNNs), although bringing new ideas[7], are unable to dynamically adjust the intensity of text-attribute interactions due to the fixed-weight fusion approach[8], and are not able to dynamically adjust the strength of text-attribute interactions in the case of data imbalance for the 'high-paying decoy', 'risky area', 'high salary', 'high risk area', etc.[9]. 'The former is a 'misleading signal' in signalling theory, and the latter is in line with the 'risky environment hypothesis' in crime hotspot theory. 'risky environment hypothesis', but existing methods lack mathematical modelling of these theories[9].

To this end, this study proposes the GAR-HGNN methodology, which translates labour market and criminological theories into computable graph features by modelling industry-pay anomaly indices (which quantify the match between job pay and industry benchmarks) and geography-industry risk density (which portrays the percentage of fraudulent jobs in a target industry within a region) to provide a theoretically-driven solution to the detection challenge. The research contributions are listed below:

- (1) Combining the anomaly index and risk density to construct a risk-orientated heterogeneous map to improve the identification of two types of typical fraud patterns;
- (2) Introducing a gated attention mechanism to dynamically adjust the fusion weights of text and attribute features to solve the context insensitivity problem of static fusion;
- (3) Designing an anomaly-sensitive weighted loss function to enhance the model's ability to learn from a few classes of samples in severely unbalanced data.

The full paper is organised as follows: section 2 reviews related work, section 3 introduces the model framework, section 4 analyses the experimental results, and section 5 concludes and suggests future directions.



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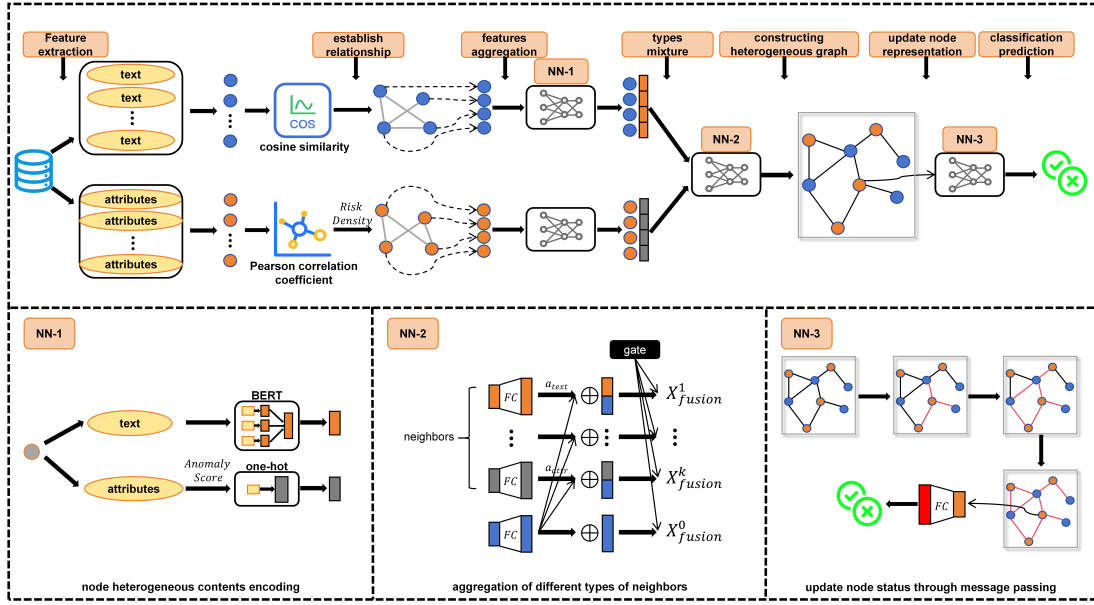


Figure 1: The overall architecture of GAR-HGNN

2 RELATED WORK

Recruitment fraud (RF) detection techniques have evolved over three generations, from machine learning, deep learning to graph neural network (GNN)-driven approaches, with each generation focusing on the core problem at a specific stage.

Early generations relied on traditional machine learning models (e.g., logistic regression, random forest) with shallow feature extraction techniques (e.g., TF-IDF, BoW). 2017 Vidros et al. achieved 91% accuracy on the EMSCAD dataset by constructing features via manual rule sets and BoW combined with a logistic regression model[10], but were weak in handling complex semantics. 2022 Amaar et al. In 2022, Amaar et al. introduced the ADASYN oversampling technique to improve the recognition of minority samples[11], but it relies on shallow features, manual rules, and insufficient generalisation. With the development of deep learning, pre-trained models based on Transformer (e.g., BERT) are used for detection. 2024 Taneja et al. proposed Fraud-BERT[12], which captures textual context by fine-tuning BERT, with an F1 value of 0.93 and an accuracy of 99% on an unbalanced dataset, but it does not incorporate multiple features and has high computational cost[13]. Akram et al. compared the performance of BERT and RoBERTa combined with SMOTE variants[14] and found that BERT+SMOBD SMOTE had a 90% recall rate. 2025 Vu et al. proposed NLP2FJD[15], which fuses text and meta-information, and is processed by CNNs to improve the comprehensiveness of the features, while ignoring the graph structure resulting in insufficient capture of hidden patterns[16].

Graph neural networks (GNNs) break through the limitations of sequence models by explicitly modelling semantics and relationships, e.g., TextGCN achieves 93.7% classification accuracy in 20NG dataset[17, 18], and a hierarchical framework combined with BERT improves long-range dependency modelling[19]. However,

traditional GNNs are difficult to deal with heterogeneous relationships, and for this reason Heterogeneous Graph Neural Networks (HGNN) have been proposed[20, 21], which construct heterogeneous graphs to model complex relationships in recruitment fraud detection by aggregating heterogeneous neighbourhood features through a random walk strategy[20]. However, the basic HGNN suffers from fixed feature fusion weights[22, 23], cannot dynamically adjust the text-attribute interaction strength[24], and does not learn enough from a few classes of samples in data imbalance scenarios (about 20:1 for real vs. fake jobs)[25, 26], resulting in poor pattern recognition accuracy for patterns such as ‘high-paying bait’.

3 GATED ATTENTION AND DUAL FEATURE ENHANCED RECRUITMENT FRAUD DETECTION BASED ON HETEROGENEOUS GRAPH NEURAL NETWORKS

The overall framework of the method covers the four closely connected core links of data preprocessing and feature extraction, homogeneous relationship subgraph construction, feature fusion and heterogeneous graph construction, model training and prediction, and the specific framework is shown in Figure 1.

3.1 Data Preprocessing and Feature Extraction

3.1.1 Data pre-processing. For numerical feature processing, based on the attribute correlation heat map(Figure 2), the attribute columns such as ‘telecommuting’ and ‘job_id’, which have low correlation with fraud, are eliminated, and the industry-salary anomaly index is constructed:

$$\text{Anomaly Score}^i = \frac{\text{salary}^i - \mu_{\text{industry}}}{\sigma_{\text{industry}}} \quad (1)$$

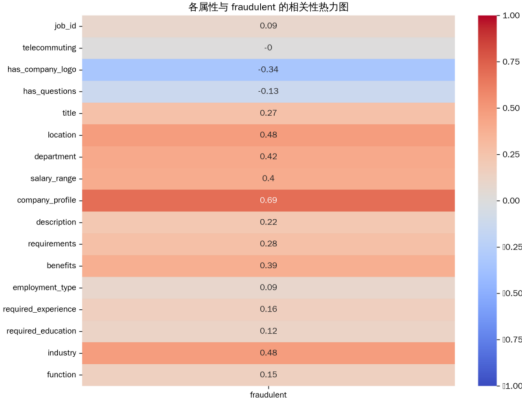


Figure 2: Heatmap of the correlation of each attribute with fraudulent

The index quantifies the risk of ‘salary - qualification’ mismatch by standardising salary deviations[27], where μ_{industry} and σ_{industry} are the industry salary mean and standard deviation. It also defines the geographic - industry risk density:

$$\text{Risk Density}(l, i) = \frac{n_{\text{fraud}}(l, i)}{n_{\text{total}}(l, i)} \quad (2)$$

Where n_{fraud} and n_{total} denote the number of fraudulent jobs in a region and the total number of jobs in the region, respectively, and the geographic-industry Risk Density reflects the risk of hiring fraud among specific regions and industries.

Text feature processing then combines multiple text fields into ‘combined_text’, cleans and normalises them, removes HTML tags, URLs, non-alphanumeric characters and stop words and converts them to lower case, and uses spaCy for word shape reduction and lexical annotation to generate the ‘pos_features’ features to better capture fraud signals[28].

3.1.2 Feature extraction. For text nodes, the pre-trained BERT model is used to extract semantic features, which are combined with the spaCy lexical annotation results to splice and generate the final text features:

$$\mathbf{x}_{\text{text-final}}^i = [\mathbf{x}_{\text{text}}^i; \mathbf{x}_{\text{pos}}^i] \quad (3)$$

where $\mathbf{x}_{\text{text}}^i$ is the BERT output vector and $\mathbf{x}_{\text{pos}}^i$ is the lexically labelled feature.

For attribute nodes a representation of the attribute node is constructed by standardising the numerical attributes, uniquely hot coding the categorical attributes, and embedding the industry-salary anomaly index as an additional feature:

$$\mathbf{x}_{\text{attr}}^i = [x_{\text{num}}^i; x_{\text{cat}}^i; \text{Anomaly Score}^i] \quad (4)$$

High anomaly index samples are labelled as high risk and given higher weights in training.

3.2 Homogeneous relationship subgraph construction

The recruitment data is modelled as a graph $G = (V, E)$. The nodes contain text nodes V_{text} (job description, etc.) and attribute nodes

V_{attr} (salary, location, etc.). Edge construction is based on data intrinsic relationships. Text similarity is calculated by cosine similarity, above a threshold $T1$ the edges are connected (the threshold calculation procedure is explained in the experimental section):

$$w_{\text{text-text}}^{ij} = \begin{cases} 1, & \text{if similarity}(v_{\text{text}}^i, v_{\text{text}}^j) > T1 \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

Based on the Pearson correlation coefficient to judge attribute correlation, above the threshold $T2$, geographic - industry risk density is introduced as a weight correction factor to enhance the edge weights of high-risk regions and improve the impact of fraud propagation paths.

$$w_{\text{attr-attr}}^{ij} = \begin{cases} 1 + \gamma \cdot \text{Risk Density}(l, i) & \text{if correlation}(v_{\text{attr}}^i, v_{\text{attr}}^j) > T2 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

3.3 Feature Fusion and Heterogeneous Map Construction

Fusion of text and attribute features using an attention mechanism to map both types of features to the same dimensional space:

$$\begin{aligned} z_{\text{text}} &= W_{\text{text}} x_{\text{text}} + b_{\text{text}}, \\ z_{\text{attr}} &= W_{\text{attr}} x_{\text{attr}} + b_{\text{attr}}, \end{aligned} \quad (7)$$

Weights are dynamically assigned through dot product score normalisation:

$$\begin{aligned} \alpha_{\text{text}} &= \frac{\exp(s_{\text{text}})}{\exp(s_{\text{text}}) + \exp(s_{\text{attr}})}, \\ \alpha_{\text{attr}} &= \frac{\exp(s_{\text{attr}})}{\exp(s_{\text{text}}) + \exp(s_{\text{attr}})} \end{aligned} \quad (8)$$

where s_{text} and s_{attr} are the attention scores, obtained by dot product calculation:

$$s_{\text{text}} = z_{\text{text}}^T z_{\text{attr}}, s_{\text{attr}} = z_{\text{attr}}^T z_{\text{text}}, \quad (9)$$

The fusion weights are dynamically adjusted by generating gating values through sigmoid functions:

$$\text{gate} = \sigma(W_{\text{gate}}[x_{\text{text}}; x_{\text{attr}}] + b_{\text{gate}}) \quad (10)$$

$$x_{\text{fusion}} = \text{gate} \cdot \alpha_{\text{text}} x_{\text{text}} + (1 - \text{gate}) \cdot \alpha_{\text{attr}} x_{\text{attr}} \quad (11)$$

Enhance the fusion efficiency of text and attribute features through dynamic weight assignment [30].

3.4 Model Training and Prediction

Updating node representations by graph convolution, aggregating neighbourhood features for message passing:

$$h_{\text{text}}^i = \sigma \left(\sum_{j \in N_{\text{text}}(i)} W_{\text{text-text}} h_{\text{text}}^j + W_{\text{text-attr}} h_{\text{attr}}^j + b_{\text{text}} \right) \quad (12)$$

$$h_{\text{attr}}^i = \sigma \left(\sum_{j \in N_{\text{attr}}(i)} W_{\text{attr-attr}} h_{\text{attr}}^j + W_{\text{attr-text}} h_{\text{text}}^j + b_{\text{attr}} \right) \quad (13)$$

The fused nodes represent the input fully connected layer and the classification results are output by Softmax.

$$y = \text{Softmax}(W_{\text{out}} h_{\text{fusion}} + b_{\text{out}}) \quad (14)$$

Design of anomaly index based weighted loss for data imbalance:

$$L = \text{NLLLoss} \cdot (1 + \alpha \cdot \sigma(\text{Anomaly} \cdot \text{Score})) \quad (15)$$

Strengthening the model's ability to learn from a few types of fraudulent samples by boosting the weight of high-risk samples.

4 EXPERIMENTS AND ANALYSIS OF RESULTS

4.1 Experimental data set

This study uses the classic Employment Scam Aegean (ESA) dataset[10], which was collected and organised by the Laboratory of Information and Communication Systems Security at the University of the Aegean, Greece. The dataset contains 18,746 job postings from 2012 to 2014, of which 17,880 are real job postings and 866 are fake job postings, which is a serious data imbalance. Each job posting in the dataset contains rich textual and meta information, which provides a comprehensive and rich data base for the study and helps to explore the characteristics of false job postings.

4.2 Experimental setup

Experimental environment: Ubuntu 22.04 system, configured with 1 RTX4090 graphics card (24GB memory), 10vCPU, 60GB RAM. The development framework is PyTorch2.2.0, based on Python programming, with imbalanced - learn and scikit - learn libraries installed for data processing and model training.

For parameter selection, text similarity and attribute similarity thresholds were determined by grid search combined with five-fold cross-validation. After maximising the F1 value on the validation set, $T1 = 0.8$ and $T2 = 0.7$ were finally selected to balance the sparsity and information richness of the graph structure. The dataset is divided into 70% training set, 20% validation set and 10% test set to ensure the reliability of model performance evaluation.

4.3 Analysis of model performance evaluation metrics

This experiment uses a variety of metrics to evaluate the performance of the model and reflect its performance in recruitment fraud detection from different dimensions:

(1) Accuracy: the percentage of samples correctly predicted by the model, formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (16)$$

Where TP is the number of correctly predicted fraudulent samples, TN is the number of correctly predicted normal samples, FP is the number of misjudged normal as fraudulent samples and FN is the number of misjudged fraudulent as normal samples. The higher the accuracy, the better the overall prediction accuracy.

(2) Precision: the proportion of samples predicted to be fraudulent that are actually fraudulent, formula:

$$Precision = \frac{TP}{TP + FP} \quad (17)$$

A high accuracy rate means that the model is reliable in predicting fraudulent samples and reduces the number of cases where normal recruitment is misjudged as fraudulent.

(3) Recall: the proportion of actual fraud samples that are correctly predicted (sensitivity), formula:

$$Recall = \frac{TP}{TP + FN} \quad (18)$$

Reflecting the model's ability to capture fraudulent samples is a key metric for recruitment fraud detection to avoid under-detection of real fraud.

(4) F1 value (F1-Score): the reconciled average of precision and recall, formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (19)$$

The balanced performance of the model on positive and negative samples is assessed comprehensively to avoid the one-sidedness of a single metric.

(5) Geometric mean (G-mean): suitable for sample imbalance scenarios, measures the combined performance of positive and negative samples, with the formula:

$$G - Mean = \sqrt{Recall \times Specificity} \quad (20)$$

It is particularly suitable for assessing the recognition ability of minority classes (fraudulent samples). (6) Specificity: the proportion of actual normal samples that are correctly predicted, formula:

$$Specificity = \frac{TN}{TN + FP} \quad (21)$$

Complementary to recall, the focus is on the accuracy of the model's judgement of normal recruitment, with the former focusing on positive class capture and the latter on negative class identification.

4.4 Contrasting Models

In order to highlight the advantages of the GAR-HGNN model proposed in this paper, we have selected several comparative models, including traditional machine learning models and deep learning models that have performed well in the field of recruitment fraud detection in recent years. The details are as follows:

Logistic regression (LR)[10]: proposed by Vidros et al. in their 2017 study, it is based on traditional statistical learning theory and predicts the category of a sample by constructing a linear regression model.

Fraud-BERT[12]: A Transformer-based context-aware recruitment fraud detection model proposed by Taneja et al. in 2024.

BERT+SMOBD SMOTE[14]: an improved model proposed by Akram et al. in 2024 that combines the pre-trained language model BERT with the density-based oversampling technique SMOBD SMOTE.

NLP2FJD[15]: an improved model proposed by Vu et al. in 2025 using deep learning based NLP techniques, which combines textual and meta-information and is further processed by Convolutional Neural Networks (CNN).

The experimental results of each model on the test set are shown in Table 1 below:

Comparison of the experimental results shows that the GAR - HGNN model outperforms or is comparable to current state-of-the-art models in several evaluation metrics, especially in key metrics such as recall rate and F1 value. This is attributed to the effective modelling of complex relationships in data by heterogeneous graph neural networks and the fusion of text and attribute features by the attention mechanism, which enables the model to better capture the features of false job postings[29], thus significantly improving the performance of fraud detection.

Table 1: Comparative experimental results table

Methodology	Accuracy	Precision	Recall	F1-score	G mean	Specificity
[10]	0.8950	0.7510	0.2820	0.4060	0.5274	0.9860
[12]	0.9817	0.8896	0.8382	0.8630	0.9134	0.9947
[14]	0.9703	0.9733	0.8984	0.8247	0.8953	0.8235
[15]	0.96	0.96	0.93	0.94	0.94	0.96
Ours	0.9896	0.9626	0.9962	0.9791	0.9919	0.9875

5 CONCLUSIONS AND FUTURE WORK

In order to solve the defects of traditional detection methods in dealing with heterogeneous data, this paper proposes the GAR-HGNN method, which achieves high-precision recruitment fraud detection on EMSCD dataset through heterogeneous graph neural network, gating attention mechanism and double feature enhancement, and provides support for platform risk control and social credit system.

Future research can be carried out in three aspects: in model optimisation, improve graph convolution operation with deep learning[30], explore better graph neural network architecture and gated attention mechanism, reduce computational consumption and improve large-scale data processing capability; in data utilisation, introduce multi-source data, such as social network, enterprise reputation, etc., to explore potential links to enrich feature representation and improve detection accuracy; in interpretability research, develop visualisation tools to show the level of feature attention through heatmap and node importance ranking to enhance user trust and safeguard job market security; In terms of interpretability research, visualisation tools are developed to show the degree of attention to model features through heatmaps and node importance ranking to enhance user trust and safeguard the security of the recruitment market.

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